ក្របខ័ណ្ឌលំហាត់ត្រៀមប្រឡងសញ្ញាបត្រមធ្យមសិក្សាទុតិយភូមិ២០២២

អាំងតេក្រាល

1. គណនាអាំងតេក្រាល៖ A= $\int_1^2 (6x-5)\,dx$ B = $\int_1^3 (3x^2-2)\,dx$ C = $\int_2^3 (4x-1-2x^2)\,dx$

D = $\int_{-1}^1 \left(\frac{x}{3}-\frac{x^2}{2}\right)dx$ E = $\int_0^1 \left(\frac{x^3}{2}-\frac{x^2}{3}+\frac{x}{4}+1\right)dx$ F = $\int_1^3 (x-2+3x^2)\,dx$ G = $\int_1^3 (2x^2-3x+1)\,dx$

H = $\int_0^2 (6x^2-3x-1)\,dx$ I = $\int_0^2 (x-1)^2\,dx$ J = $\int_1^2 (2x-1)^2\,dx$ K = $\int_0^1 (x^2+1)^2\,dx$

L = $\int_0^1 (2x^2+3)^2\,dx$ M = $\int_2^3 x(x-3)\,dx$ N = $\int_{-1}^2 (x-1)(x+2)\,dx$ O = $\int_0^1 x(x+1)^2\,dx$

2. គណនាអាំងតេក្រាល៖ A = $\int_1^2 \left(\frac{1}{x^2}+\frac{1}{x}\right)dx$ B = $\int_1^2 \left(x-\frac{1}{x}+\frac{2}{x^2}\right)dx$ C = $\int_1^2 \left(\frac{2+x-2x^2}{x}\right)dx$

D = $\int_1^2 \left(\frac{x^2+2x-3}{x^2}\right)dx$ E = $\int_1^e \left(\frac{1}{x}-\frac{e}{x^2}\right)dx$ F = $\int_0^1 \left(2x+1-\frac{1}{x+1}\right)dx$ G = $\int_2^3 \left(3-x+\frac{2}{1-x}\right)dx$

H = $\int_0^1 \left(x+e-\frac{2}{x+e}\right)dx$ I = $\int_1^2 \left(x-2+\frac{3}{2x-1}\right)dx$ J = $\int_0^1 \left(x^2-\frac{2}{3x-1}\right)dx$ K = $\int_0^1 \left(\frac{x}{2}+\frac{4}{3-2x}\right)dx$

L = $\int_0^1 \left(x+\frac{x}{x^2+1}\right)dx$ M = $\int_{-1}^0 \left(x-\frac{4x}{x^2+3}\right)dx$ N = $\int_1^2 \left(2x-1+\frac{2x}{x^2-2}\right)dx$ O = $\int_0^1 \left(2+\frac{2}{(x+1)^2}\right)dx$

P = $\int_2^3 \left(2x-\frac{4}{(x-1)^2}\right)dx$ Q = $\int_0^1 \left(x-\frac{3}{(2x+1)^2}\right)dx$ R = $\int_{-1}^0 \left(\frac{2}{2x+3}-\frac{1}{(2x+3)^2}\right)dx$ S = $\int_2^3 \left(\frac{2}{x-1}-\frac{1}{(x-1)^2}\right)dx$

3. គណនាអាំងតេក្រាល៖ A = $\int_0^{\ln 2} (e^x+1)\,dx$ B = $\int_0^{\ln 3} (e^x-4)\,dx$ C = $\int_0^{\ln 2} (e^x-2x+1)\,dx$

D = $\int_0^1 (3e^{2x}+e^{-2x})\,dx$ E = $\int_0^1 (2-4e^{3x})\,dx$ F = $\int_0^{\ln 2} (e^x+1)^2\,dx$ G = $\int_0^{\ln 3} (2e^x-1)^2\,dx$

H = $\int_0^{\ln 2} e^x(2e^x-e^{-x})\,dx$ I = $\int_0^{\ln 2} e^{2x}(e^x-2e^{-2x})\,dx$ J = $\int_0^{\ln 2} \left(\frac{e^x}{e^x+1}\right)dx$ K = $\int_{\ln 2}^{\ln 3} \left(\frac{2e^x}{e^x-1}\right)dx$

L = $\int_0^{\ln 3} \left(e^x-\frac{e^x}{e^x+1}\right)dx$ M = $\int_0^{\ln 2} \left(e^x+1-\frac{e^x}{e^x+3}\right)dx$ N = $\int_0^{\ln 2} \left(e^x+\frac{2e^x}{2e^x+1}\right)dx$ O = $\int_0^{\ln 2} \left(2-\frac{4e^x}{1-2e^x}\right)dx$

P = $\int_0^{\ln 2} e^x\left(\frac{3}{e^x}-\frac{8}{e^x+2}\right)dx$ Q = $\int_0^{\ln 2} \frac{e^x}{(e^x+1)^2}\,dx$ R = $\int_{\ln 3}^{\ln 5} \frac{4e^x}{(e^x-2)^2}\,dx$ S = $\int_0^{\ln 3} \frac{3e^x}{(3-2e^x)^2}\,dx$

4. គណនាអាំងតេក្រាល៖ A = $\int_0^{\frac{\pi}{2}} (2+\cos x)\,dx$ B = $\int_0^{\frac{\pi}{3}} (1+2\sin x)\,dx$ C = $\int_{\frac{\pi}{2}}^{\pi} (2\cos x-\sin x)\,dx$

D = $\int_0^{\frac{\pi}{4}} (2+\cos 2x)\,dx$ E = $\int_0^{\frac{\pi}{4}} (1-\sin 2x)\,dx$ F = $\int_0^{\frac{\pi}{6}} (\sin 2x+\cos 3x)\,dx$ G = $\int_0^{\frac{\pi}{12}} (4\cos 2x-\sin 4x)\,dx$

H = $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \left(\frac{1}{3}\cos x+\frac{1}{2}\sin 2x\right)dx$ I = $\int_0^{\frac{\pi}{2}} \cos x\sin^2 x\,dx$ J = $\int_0^{\frac{\pi}{3}} \cos^3 x\sin x\,dx$ K = $\int_0^{\frac{\pi}{4}} 2\sin^4 x\cos x\,dx$

L = $\int_0^{\frac{\pi}{6}} \sin x(1-\cos^2 x)\,dx$ M = $\int_0^{\frac{\pi}{6}} \cos x(3+2\sin^3 x)\,dx$ N = $\int_0^{\frac{\pi}{4}} \sin^2 x\,dx$ O = $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} (2\cos^2 x-1)\,dx$

P= $\int_0^{\frac{\pi}{12}} (1-4\sin^2 x)\,dx$ Q = $\int_0^{\frac{\pi}{2}} \cos\left(x+\frac{\pi}{2}\right)dx$ R = $\int_0^{\frac{\pi}{4}} 2\sin\left(x+\frac{\pi}{4}\right)dx$ S = $\int_0^{\frac{\pi}{3}} \sin\left(2x+\frac{\pi}{3}\right)dx$

T = $\int_0^{\frac{\pi}{6}} 2\cos\left(3x-\frac{\pi}{3}\right)dx$ U = $\int_0^{\frac{\pi}{4}} 2\sin\left(3x-\frac{\pi}{4}\right)dx$ U = $\int_0^{\frac{\pi}{4}} \left[\sin^2 x\cos x+\cos\left(x+\frac{\pi}{4}\right)\right]dx$

V = $\int_0^{\frac{\pi}{6}} \left[2\cos^3 x\sin x-3\sin\left(2x+\frac{2\pi}{3}\right)\right]dx$ W = $\int_0^{\frac{\pi}{4}} \left(2\cos^2 x-\sin\left(3x+\frac{\pi}{4}\right)\right)dx$

5. កំណត់តម្លៃ a និង b ដើម្បីឱ្យ $\frac{2x+1}{x+1}=a+\frac{b}{x+1}$ ចំពោះ $x\neq -1$ ។រួចគណនាតម្លៃ $\int_0^1\left(\frac{2x+1}{x+1}\right)dx$ ។

6. កំណត់តម្លៃ a និង b ដើម្បីឱ្យ $\frac{-3x-2}{x-1}=a+\frac{b}{x-1}$ ចំពោះ $x\neq 1$ ។រួចគណនាតម្លៃ $\int_0^2\left(\frac{-3x-2}{x-1}\right)dx$ ។

7. កំណត់តម្លៃ a និង b ដើម្បីឱ្យ $\frac{3x+1}{x+3}=a+\frac{b}{x+3}$ ចំពោះ $x\neq -3$ ។រួចគណនាតម្លៃ $\int_0^1\left(\frac{3x+1}{x+3}\right)dx$ ។

8. រកតម្លៃ $a\,;b$ និង c ដើម្បីឱ្យ $\frac{3x^2+4x-1}{2-x}=ax+b+\frac{c}{2-x}$ ចំពោះ $x\neq 2$ ។គណនាតម្លៃ $\int_0^1\left(\frac{3x^2+4x-1}{2-x}\right)dx$ ។

9. គេមានអនុគមន៍ $f(x)=\frac{2x^2-5x+2}{3x}\,;x\neq 0$ ។កំណត់តម្លៃ $a;b$ និង c ដើម្បីឱ្យ $f(x)=ax+b+\frac{c}{3x}$ ។

គណនាអាំងតេក្រាល $I=\int_1^2\left(\frac{2x^2-5x+2}{3x}\right)dx$ ។

10. គេមាន $f(x)=\frac{x^2-x-2}{x-1}\,;x\neq 1$ ។រកតម្លៃ $a;b$ និង c ដើម្បីឱ្យ $f(x)=ax+b+\frac{c}{x-1}$ ។គណនា $\int_0^2 f(x)dx$ ។

11. គេមានអនុគមន៍ $f(x)=\frac{1}{x^2-4}$ ។ រកតម្លៃ a និង b ដើម្បីឱ្យ $f(x)=\frac{a}{x-2}+\frac{b}{x+2}$ ។គណនា $\int_0^1\frac{1}{x^2-4}dx$ ។

12. បង្ហាញថាចំពោះ $x\in\mathbb{R}-\{-2\}$ គេបាន $\frac{x^2+x-1}{x+2}=x-1+\frac{1}{x+2}$ ។រួចគណនា $\int_{-1}^1\frac{x^2+x-1}{x+2}dx$ ។

13. បង្ហាញថាចំពោះ $x\in\mathbb{R}-\{2\}$ គេបាន $\frac{x^2-2x-3}{x-2}=x+\frac{1}{x-2}$ ។រួចគណនា $\int_{-1}^1\frac{x^2-2x-3}{x-2}dx$ ។

14. បង្ហាញថាចំពោះ $x\in\mathbb{R}-\{1\}$ គេបាន $\frac{x^2-2x+2}{x-1}=x-1+\frac{1}{x-1}$ ។រួចគណនា $\int_0^2\left(\frac{x^2-2x+2}{x-1}\right)dx$ ។

15. បង្ហាញថាចំពោះ $x\in\mathbb{R}-\{2\}$ គេបាន $\frac{x^2-3x+3}{2-x}=-x+1+\frac{1}{2-x}$ ។រួចគណនា $I=\int_3^4\left(\frac{x^2-3x+3}{2-x}\right)dx$ ។

16. (BacII,2015) គណនាអាំងតេក្រាល: ក. $I=\int_1^2\left(\frac{x^2}{3}-\frac{x}{2}+3\right)dx$

ខ. គេឲ្យអនុគមន៍ $f(x)=-\frac{2-x}{(x-1)^2}$ ។ បង្ហាញថា $f(x)=-\frac{1}{(x-1)^2}+\frac{1}{x-1}$ ។ គណនា $K=\int_{-1}^0 f(x)dx$ ។

17. (BacII,2016) គណនាអាំងតេក្រាល $I=\int_0^2\left(6x^2-3x-1\right)dx$, $J=\int_0^{\frac{\pi}{4}}\left(1-2\sin^2 x\right)dx$ ។ គេមាន f កំណត់លើ $\mathbb{R}*$ ដោយ $f(x)=-2\left(\frac{x+1}{x^2}\right)$ ។ បង្ហាញថា $f(x)=-\frac{2}{x}-\frac{2}{x^2}$ ។ គណនា $K=\int_1^e f(x)dx$ ។ ($\ln e=1$)

18. (BacII,2017) គណនាអាំងតេក្រាល $I=\int_1^3\left(x-2+3x^2\right)dx$, $J=\int_0^{\frac{\pi}{4}}\left(\sin 2x-\cos x\right)dx$, $K=\int_0^1\frac{x^3+(x+1)^2}{x^2+1}dx$ ។

ដើម្បីគណនា K យើងត្រូវបង្ហាញថា $\frac{x^3+(x+1)^2}{x^2+1}=x+1+\frac{x}{x^2+1}$ ។

19. (BacII,2018) គណនាអាំងតេក្រាល $I=\int_1^2\left(2-x+x^2\right)dx$, $J=\int_0^{\frac{\pi}{4}}\left(\cos 2x-\frac{1}{4}\cos 4x\right)dx$, $K=\int_2^3\left(3x-2+\frac{1}{x-1}\right)dx$

20. (BacII,2019) គណនាអាំងតេក្រាល $I=\int_0^1\left(x^2+1\right)^2dx$; $J=\int_0^{\ln 6}\left(e^x-1\right)dx$; $K=\int_0^{\frac{\pi}{4}}\left[\sin\left(3x+\frac{\pi}{4}\right)+\sin^4 x\cos x\right]dx$

21. (BacII,2021) គណនាអាំងតេក្រាល $I=\int_0^1\left(3x^2-2x+1\right)dx$; $J=\int_0^1\left(\frac{e^x}{e^x+1}\right)dx$; $K=\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\sin^3 x\cos x\right)dx$